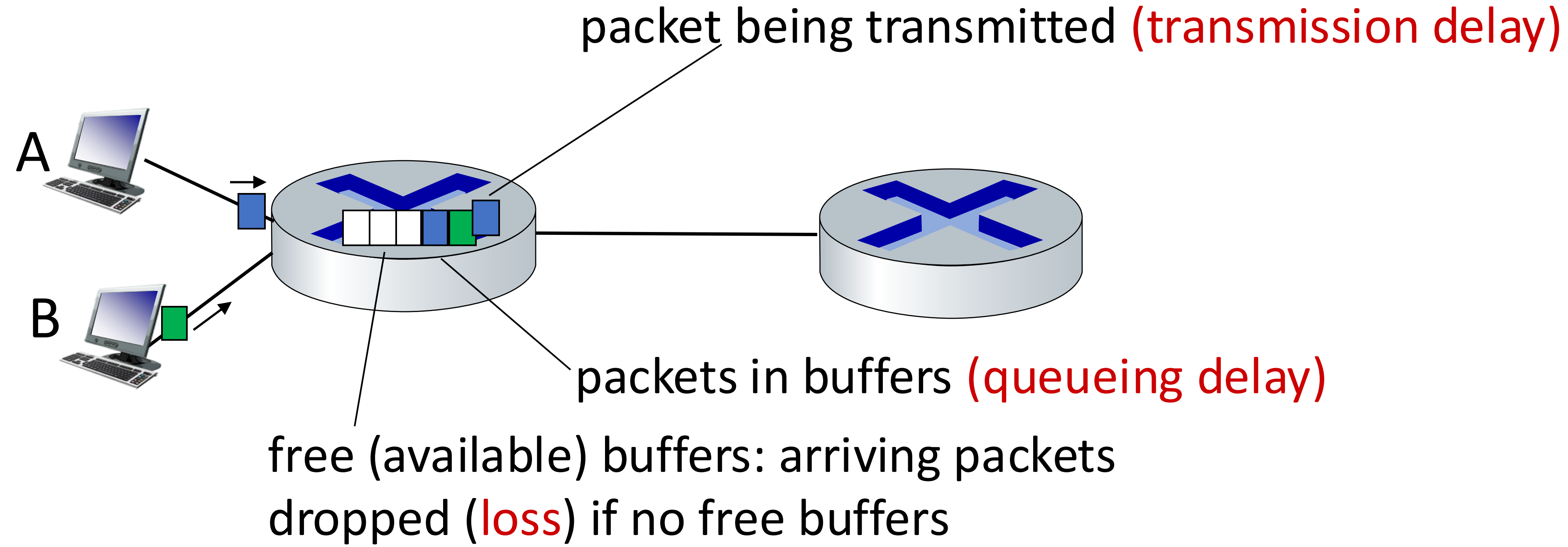
# Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History

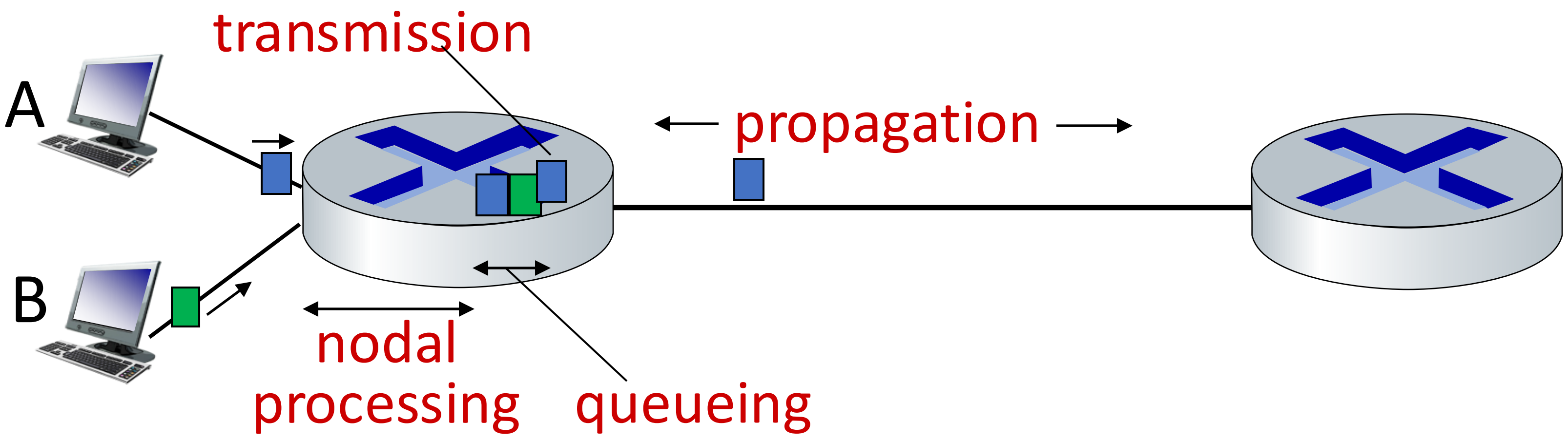
# How do packet loss and delay occur?

packets *queue* in router buffers

- packets queue, wait for turn
- arrival rate to link (temporarily) exceeds output link capacity: packet loss



# Packet delay: four sources (tilsammen $d_{sum}$)




$$d_{sum} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

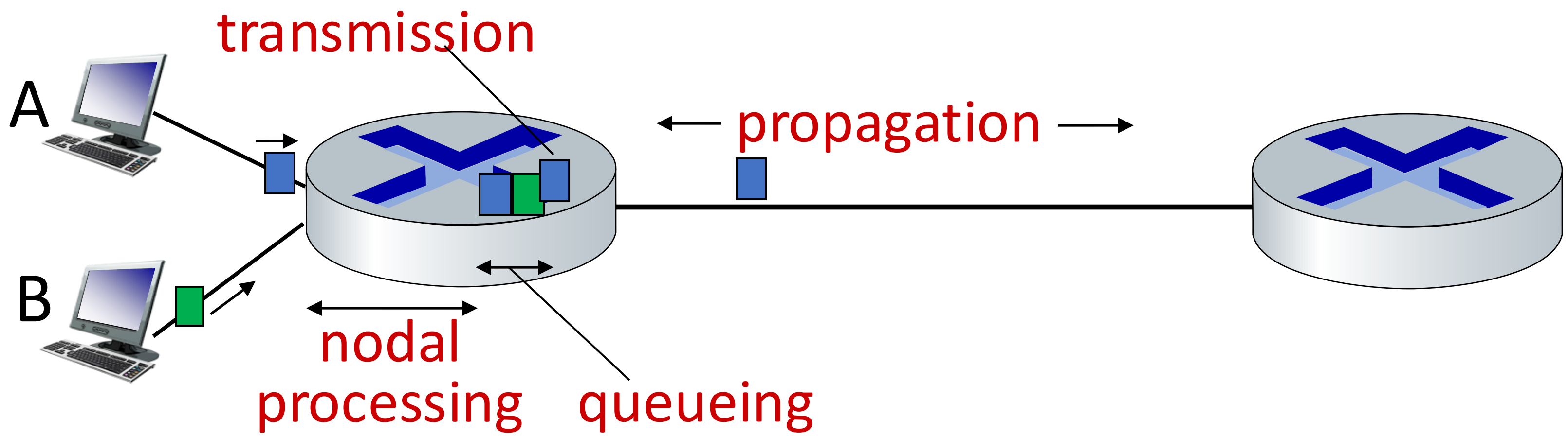
**$d_{proc}$: nodal processing**

- check bit errors
- determine output link
- typically < msec

**$d_{queue}$: queueing delay**

- time waiting at output link for transmission
- depends on congestion level of router

# Packet delay: four sources (tilsammen $d_{sum}$)




$$d_{sum} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

$d_{trans}$: transmission delay:

- $L$: packet length (bits)
- $R$: link *transmission rate (bps)*
- $d_{trans} = L/R$

$d_{prop}$: propagation delay:

- $d$: length of physical link
- $s$: propagation speed (~2x$10^8$ m/sec)
- $d_{prop} = d/s$

$d_{trans}$ and $d_{prop}$
*very* different

* Check out the online interactive exercises:
http://gaia.cs.umass.edu/kurose_ross

# La oss se litt på $d_{prop} = d/s$ og $d_{trans} = L/R$

Dette kalles fiberens refraksjonsindeks eller brytningsindeks, som er om lag 1,47 i en typisk singelmodusfiber. Dette gir en lyshastighet på **300 000 kilometer per sekund dividert med 1,47**; det vil si om lag 200 000 kilometer per sekund. Dette er om lag samme signalhastighet som i kobberkabler.

$d_{prop} = d/s$

$d_{trans} = L/R$

# La oss se litt på $d_{prop} = d/s$ og $d_{trans} = L/R$

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$d_{prop} = d/s$

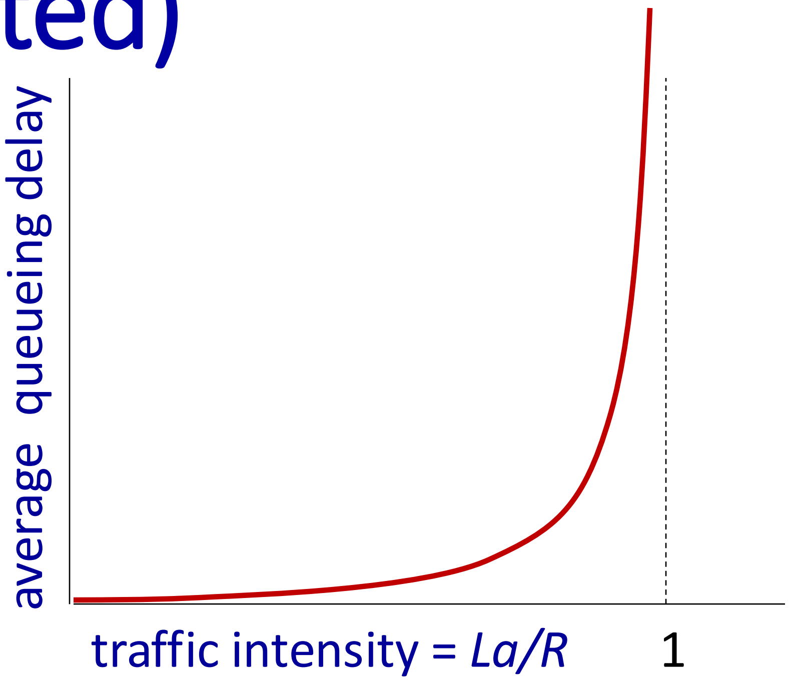
d = 540km (Trondheim – Oslo)
s = 200 000 000 m/s
**d prop = d/s = 2,7ms**


$d_{trans} = L/R$

L = 1500byte = 1500x8 bits = 12.000 bits
R = 1Gbps = 1.000.000.000 bits/sec
**d trans = L/R = 0,012ms**

# Packet queueing delay (revisited)

- *R:* link bandwidth (bps)
- *L:* packet length (bits)
- *a:* average packet arrival rate



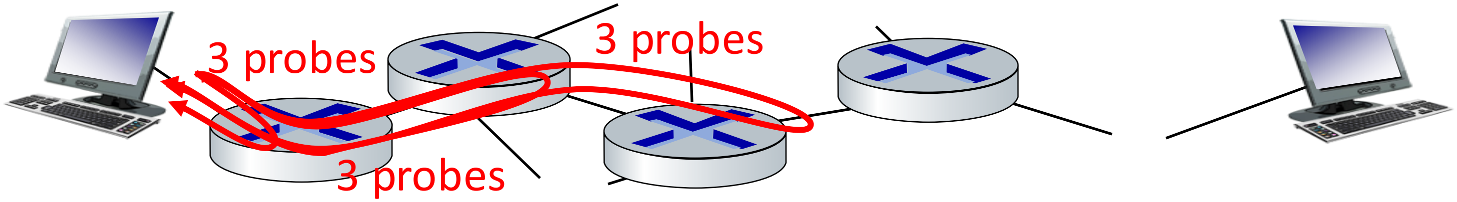

- *La/R* ~ 0: avg. queueing delay small
- *La/R* -> 1: avg. queueing delay large
- *La/R* > 1: more "work" arriving is more than can be serviced - average delay infinite!

*La/R* ~ 0

*La/R -> 1*

# "Real" Internet delays and routes

- what do "real" Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all *i:*
  - sends three packets that will reach router *i* on path towards destination (with time-to-live field value of *i*)
  - router *i* will return packets to sender
  - sender measures time interval between transmission and reply



# Real Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

trans-oceanic link

looks like delays *decrease*! Why?

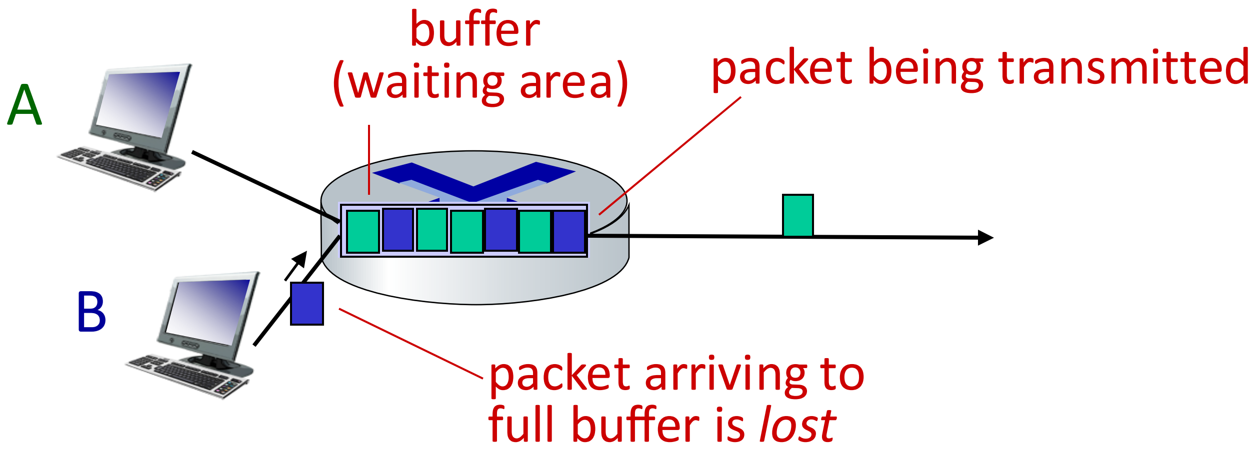
* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

# Når vi måler selv…eks vha nettfart.no


# Packet loss

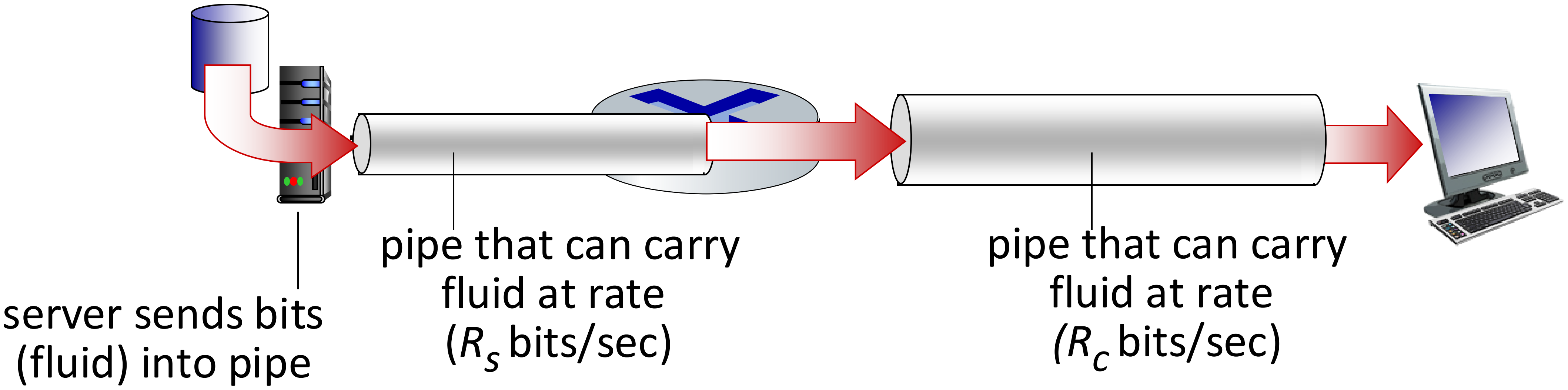
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all




* Check out the Java applet for an interactive animation on queuing and loss

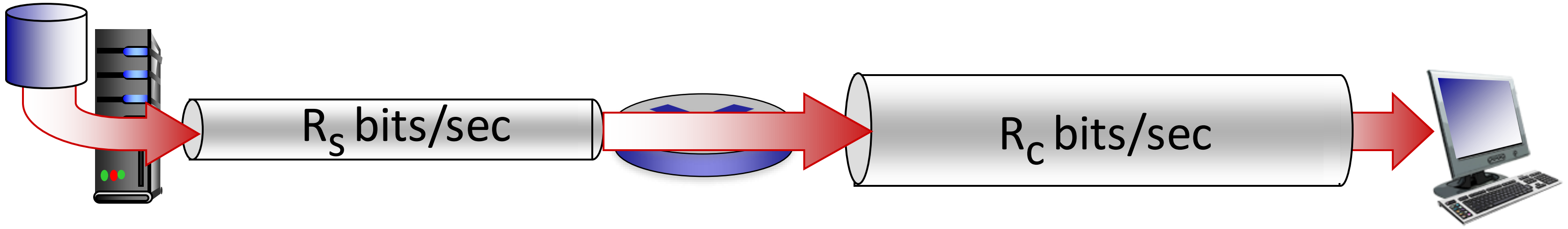
# Throughput

- *throughput:* rate (bits/time unit) at which bits are being sent from sender to receiver
  - *instantaneous:* rate at given point in time
  - *average:* rate over longer period of time

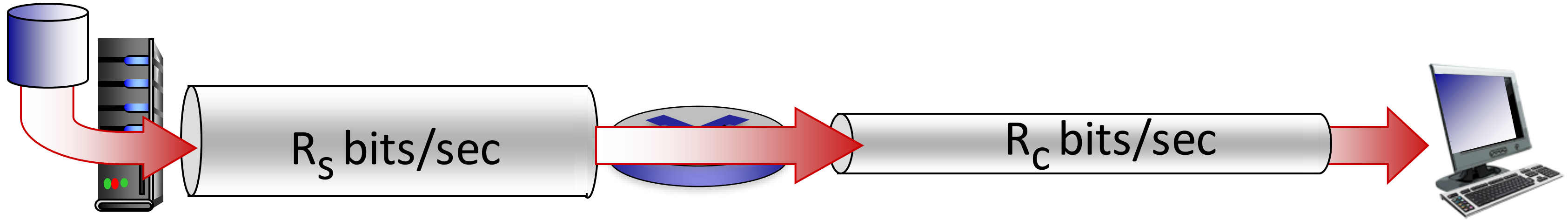


# Throughput

$R_s < R_c$ What is average end-end (potential) throughput?



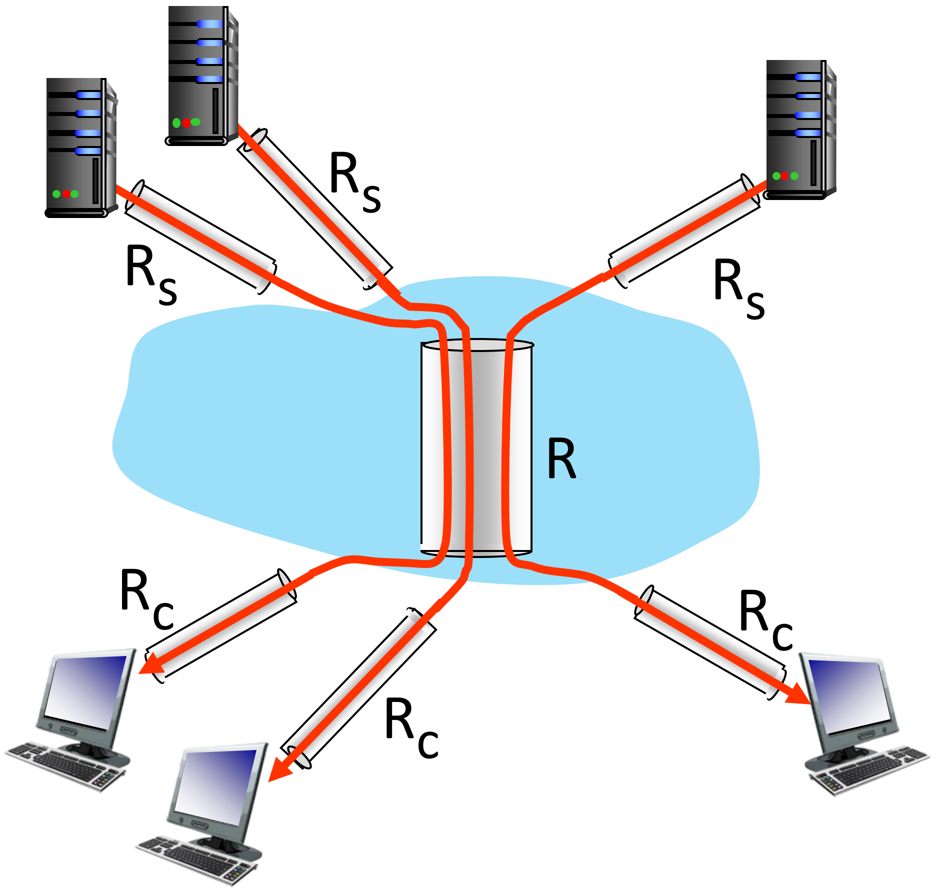

$R_s > R_c$ What is average end-end (potential) throughput?




*bottleneck link*

link on end-end path that constrains end-end throughput

# Throughput: network scenario




10 connections (fairly) share backbone bottleneck link *R* bits/sec

- per-connection end-end throughput: *min($R_c$,$R_s$,R/10)*
- in practice: $R_c$ or $R_s$ is often bottleneck

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/

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# Network security

- field of network security:
  - how bad guys can attack computer networks
  - how we can defend networks against attacks
  - how to design architectures that are immune to attacks
- Internet not originally designed with (much) security in mind
  - *original vision:* "a group of mutually trusting users attached to a transparent network" ☺
  - Internet protocol designers playing "catch-up"
  - security considerations in all layers!

# Bad guys: malware

- malware can get in host from:
  - *virus:* self-replicating infection by receiving/executing object (e.g., e-mail attachment)
  - *worm:* self-replicating infection by passively receiving object that gets itself executed
- spyware malware can record keystrokes, web sites visited, upload info to collection site
- infected host can be enrolled in botnet, used for spam or distributed denial of service (DDoS) attacks

bard.google.com/u/1/chat/507873c4c742e518

Trondheim kommune | Bluegarden Portal | Intranett - Trondhei... | Bilbooking | Min konto | Kommunedirektøre... | IT-tjenesten | Assets | ServiceNow | IT-Tjenestens aktivit... | NTNU

Bard Eksperiment

Se de siste oppdateringene i hjelpesenteret for personvern

Ny chat

**2024**

Nylige

- Sjanse for botnet-infeksjon
- Parlamentarisme i kommuner
- The World in Flux
- Vis mer

hvor stor andel av PC i hele verden er del av botnet ?

Se andre utkast

Det er vanskelig å si nøyaktig hvor stor andel av PC-er i hele verden som er del av et botnet. Men ifølge en rapport fra Cybersecurity Ventures vil det være 1,35 milliarder infiserte enheter i verden innen 2025. Dette utgjør omtrent 20 % av alle PC-er i verden.

Rapporten sier at det er en økende tendens til at botnets blir mer sofistikerte. De blir nå brukt til en rekke forskjellige formål, inkludert DDoS-angrep, phishing og datatyveri.

Det er flere ting som kan gjøres for å redusere risikoen for at PC-en din blir infisert med et botnet. Her er noen tips:

- Bruk et sterkt passord for PC-en din.
- Aktiver tofaktorautentisering.
- Oppdater programvaren din regelmessig.
- Bruk sikkerhetsprogramvare som beskytter mot malware.
- Vær forsiktig med hvilke nettsteder du besøker.

Skriv en forespørsel her

Saupstad, Norge

Gemini Advanced

**2025**

hvor stor andel av PC'er i verden er sannsynligvis del av et botnet ?

Det er vanskelig å gi et eksakt tall på hvor stor andel av PC-er i verden som er del av et botnet. Dette tallet varierer kontinuerlig, og cyberkriminelle jobber stadig med å utvikle nye metoder for å infisere maskiner.

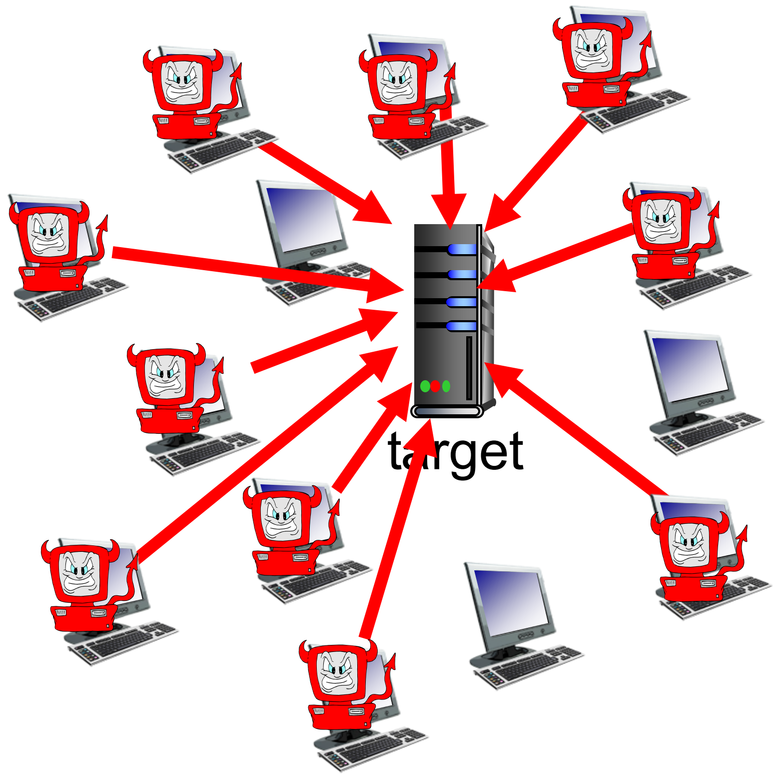
Imidlertid gir tidligere rapporter og analyser noen indikasjoner:

- **Tidligere estimater:** Noen rapporter har antydet at så mye som **25% av alle internettilkoblede datamaskiner** kan være en del av et botnet. Dette tallet er imidlertid usikkert og kan være høyere eller lavere.
- **Faktorer som påvirker antallet:** Utbredelsen av botnett påvirkes av en rekke faktorer, inkludert:
  - **Sikkerhetsbevissthet blant brukere:** Brukere med god sikkerhetspraksis, som å oppdatere programvare og unngå mistenkelige lenker og vedlegg, er mindre utsatt for infeksjoner.
  - **Sikkerhetstiltak fra programvareleverandører:** Programvareselskaper jobber kontinuerlig

# Bad guys: denial of service

*Denial of Service (DoS):* attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

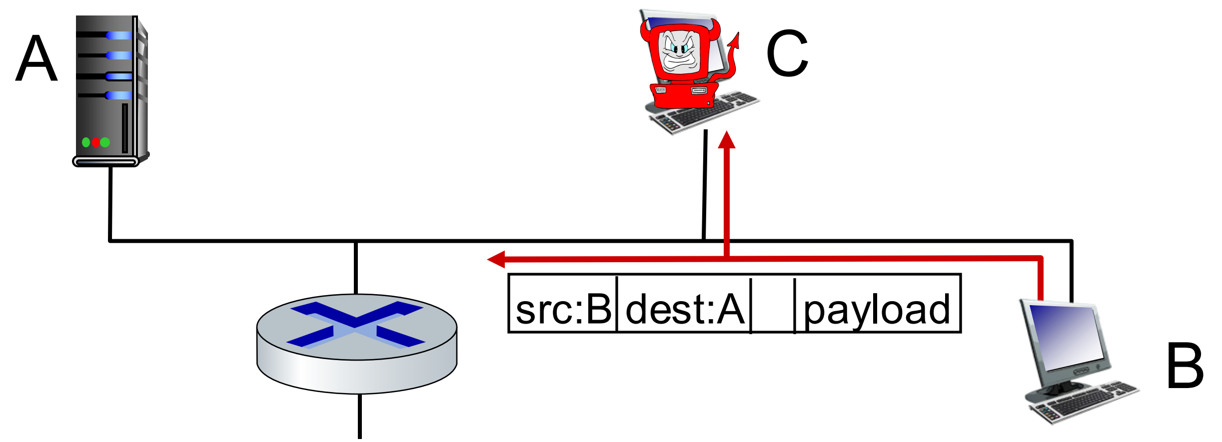
1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



# Bad guys: packet interception

*packet "sniffing":*

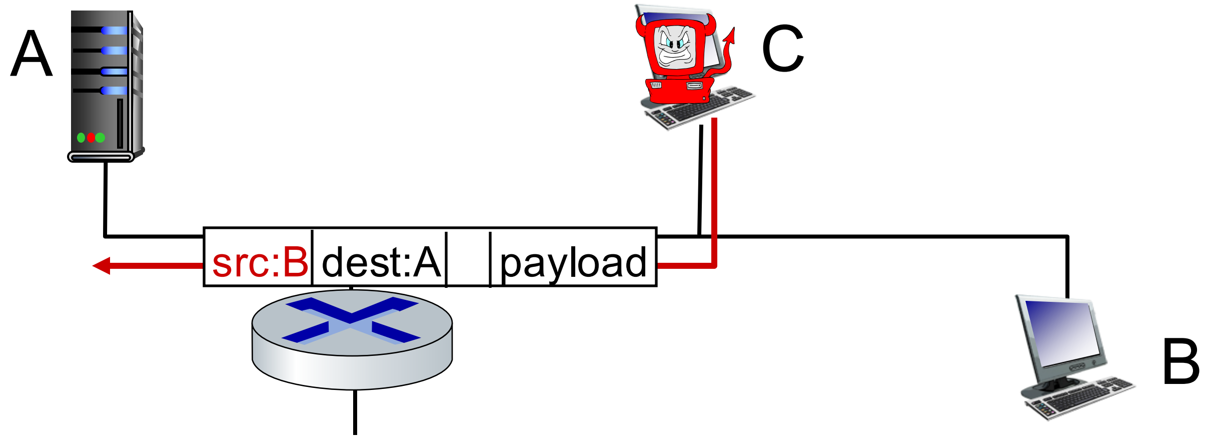
- broadcast media (shared Ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by




Wireshark software used for our end-of-chapter labs is a (free) packet-sniffer

# Bad guys: fake identity

*IP spoofing:* send packet with false source address




*… lots more on security (throughout, Chapter 8)*

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# Protocol "layers" and reference models

*Networks are complex, with many "pieces":*

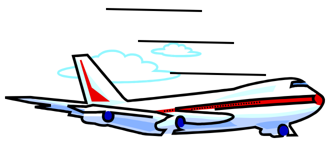
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

*Question:*

is there any hope of *organizing* structure of network?

.... or at least our *discussion* of networks?

# Example: organization of air travel

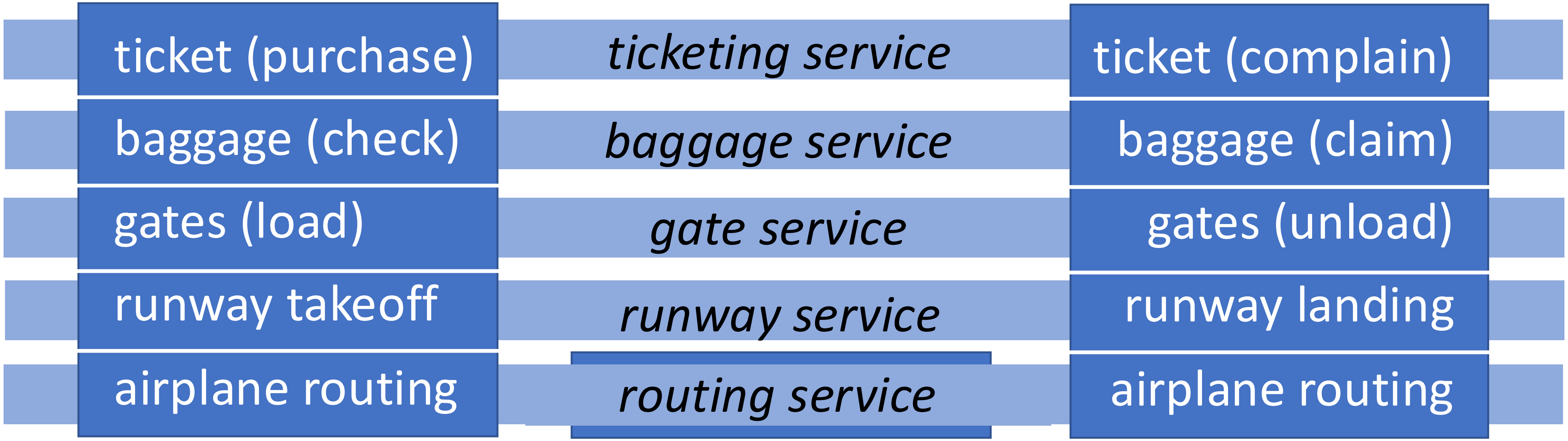


| | |
|---|---|
| ticket (purchase) | ticket (complain) |
| baggage (check) | baggage (claim) |
| gates (load) | gates (unload) |
| runway takeoff | runway landing |
| airplane routing | airplane routing |

airplane routing

airline travel: a series of steps, involving many services

# Example: organization of air travel




*layers:* each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

*Q: describe in words the service provided in each layer above*

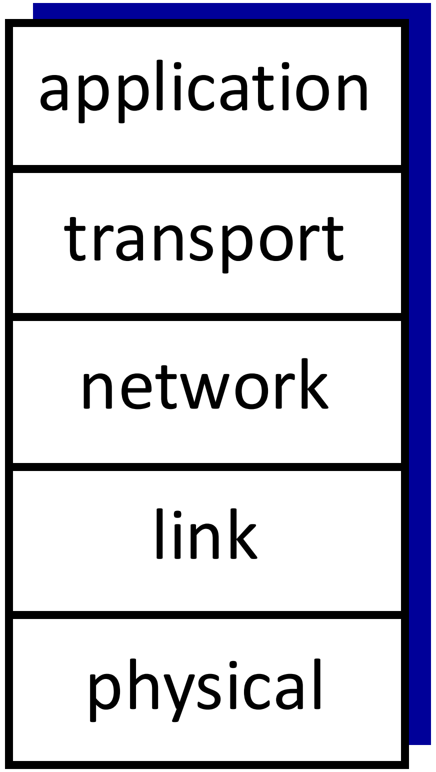
# Why layering?

dealing with complex systems:

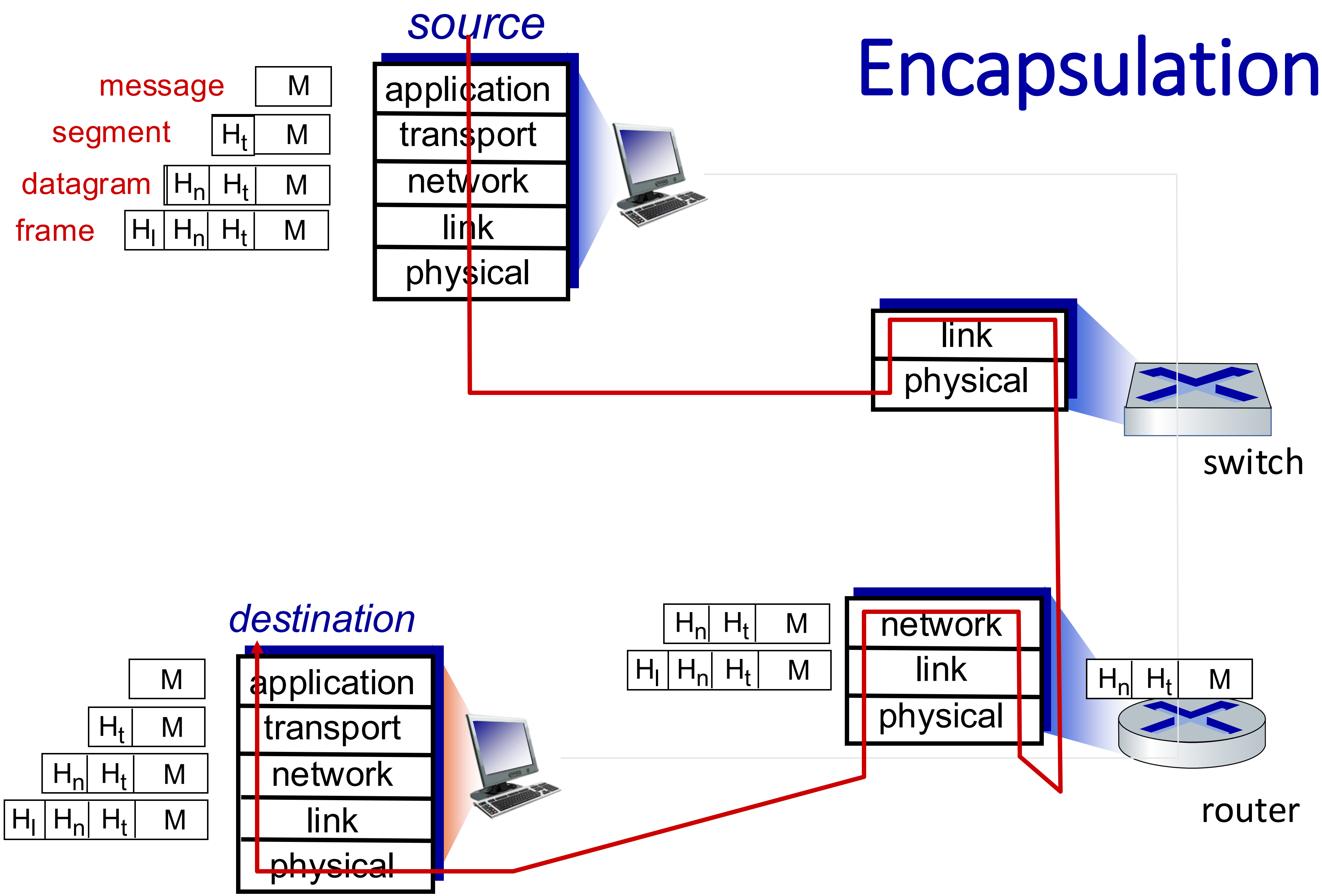
- explicit structure allows identification, relationship of complex system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change in layer's service *implementation*: transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system
- layering considered harmful?
- layering in other complex systems?

# Internet protocol stack

- *application:* supporting network applications
  - IMAP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits "on the wire"



# Encapsulation



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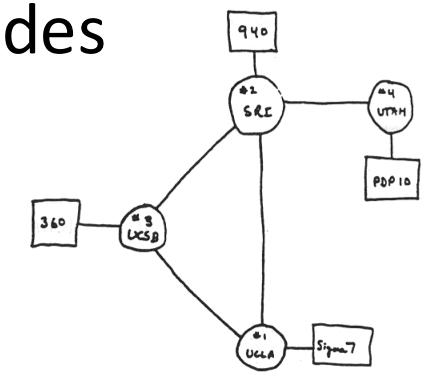
# Internet history

*1961-1972: Early packet-switching principles*

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational
- 1972:
  - ARPAnet public demo
  - NCP (Network Control Protocol) first host-host protocol
  - first e-mail program
  - ARPAnet has 15 nodes




THE ARPA NETWORK

# Internet history

## *1972-1980: Internetworking, new and proprietary nets*

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- late70's: proprietary architectures: DECnet, SNA, XNA
- late 70's: switching fixed length packets (ATM precursor)
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best-effort service model
- stateless routing
- decentralized control

define today's Internet architecture

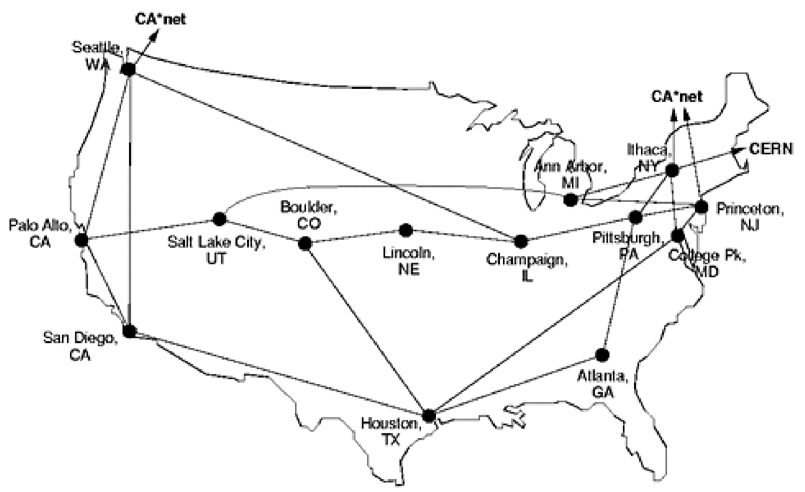
# Internet history

*1980-1990: new protocols, a proliferation of networks*

- 1983: deployment of TCP/IP
- 1982: smtp e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: ftp protocol defined
- 1988: TCP congestion control

- new national networks: CSnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks

NSFNET T1 Network 1991



# Internet history

*1990, 2000s: commercialization, the Web, new applications*

- early 1990s: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
  - hypertext [Bush 1945, Nelson 1960's]
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, later Netscape
  - late 1990s: commercialization of the Web

late 1990s – 2000s:

- more killer apps: instant messaging, P2P file sharing
- network security to forefront
- est. 50 million host, 100 million+ users
- backbone links running at Gbps

# Internet history

*2005-present: more new applications, Internet is "everywhere"*

- ~18B devices attached to Internet (2017)
  - rise of smartphones (iPhone: 2007)
- aggressive deployment of broadband access
- increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- emergence of online social networks:
- Facebook: ~ 2.5 billion users
- service providers (Google, FB, Microsoft) create their own networks
  - bypass commercial Internet to connect "close" to end user, providing "instantaneous" access to search, video content, …
- enterprises run their services in "cloud" (e.g., Amazon Web Services, Microsoft Azure)

# Chapter 1: summary

*We've covered a "ton" of material!*

- Internet overview
- what's a protocol?
- network edge, access network, core
  - packet-switching versus circuit-switching
  - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security
- history

*You now have:*

- context, overview, vocabulary, "feel" of networking
- more depth, detail, *and fun* to follow*!*

# Warriors of the net (1999)


[Ericsson Medialab]